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Experiment No. 06

Title: Simulation of Memory Management and Deadlock Handling Techniques Using CPU-OS Simulator

Objective:

- To simulate memory management techniques including virtual memory and paging using a CPU-OS simulator.
- To analyze the effects of paging and demand paging on memory allocation and process execution.
- To create and observe deadlock conditions using multiple processes and resource allocation.
- To implement and study deadlock detection, prevention, and recovery techniques.
- To examine process behavior, memory usage, and resource allocation through simulation tools.

Software: CPU-OS Simulator (YASMIN Simulator)

Theory:

1. Memory Management, Virtual Memory, and Paging

Memory management is an important function of an operating system that controls the allocation and use of main memory. It ensures efficient utilization of memory while allowing multiple processes to execute simultaneously.

 Virtual Memory

Virtual memory is a technique that allows processes to run even if they are larger than the available physical memory. It uses secondary storage to extend memory and loads only required parts of a process into RAM. This improves system performance and supports multitasking.

 Paging

Paging divides memory into fixed-size blocks called pages (logical memory) and frames (physical memory). It allows non-contiguous memory allocation, thereby eliminating external fragmentation and improving memory utilization.

 Demand Paging

In demand paging, pages are loaded into memory only when required. If a page is not present, a page fault occurs and the system loads the page from disk. This reduces memory usage but may increase overhead due to page faults.

2. Deadlocks in Operating Systems

A deadlock is a situation where two or more processes are unable to proceed because each is waiting for a resource held by another.

 Necessary Conditions

Deadlock occurs when the following conditions are present:

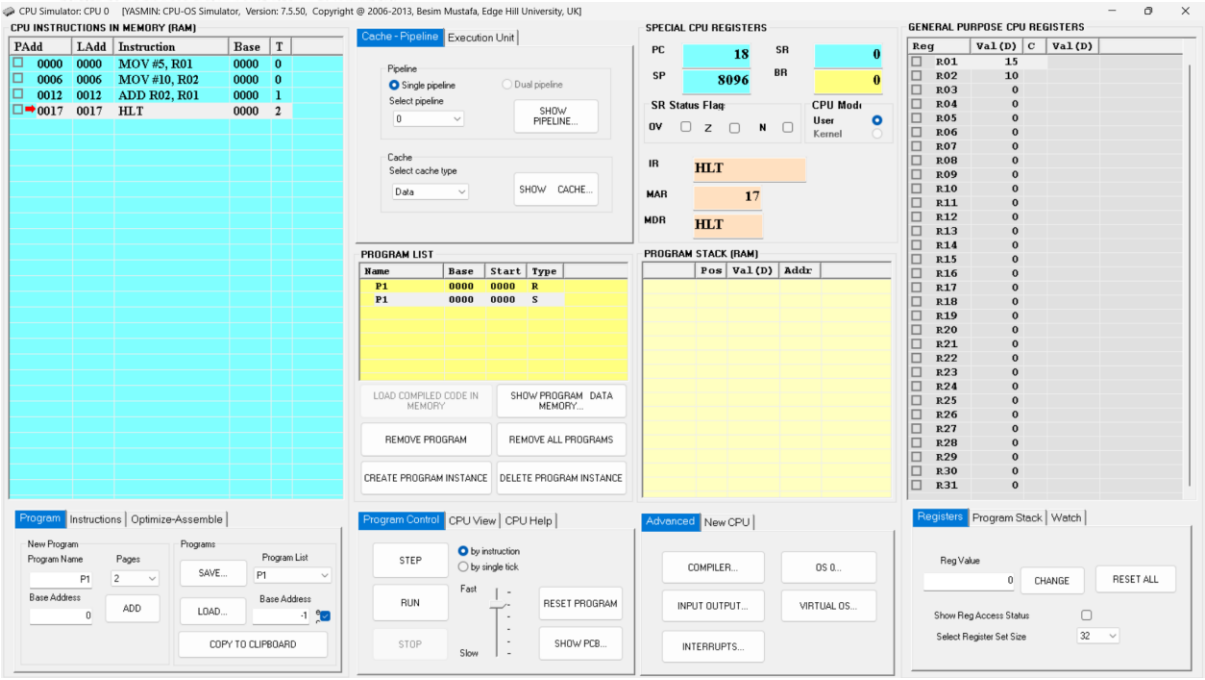
- ✧ Mutual exclusion
- ✧ Hold and wait
- ✧ No preemption
- ✧ Circular wait

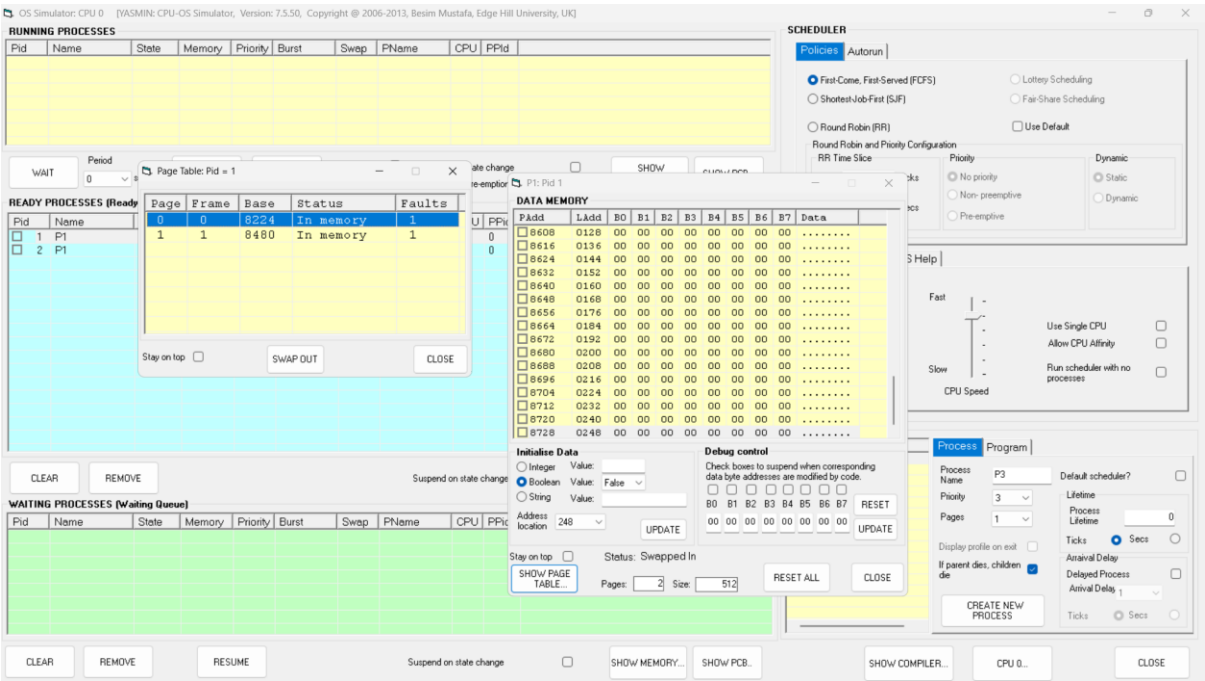
 Resource Allocation Graph: It is used to represent processes and resources. A cycle in the graph indicates a deadlock condition.

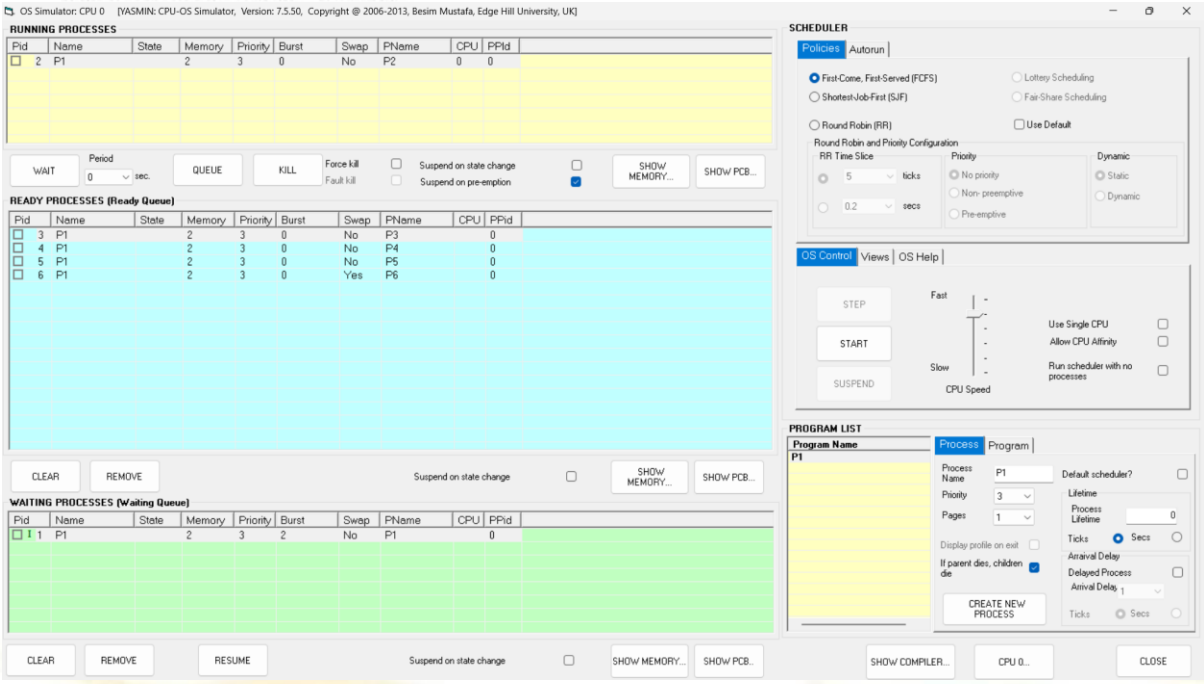
 Deadlock Handling Techniques:

- ✧ Detection: Identifying deadlocks using system analysis
- ✧ Recovery: Resolving deadlocks by releasing resources or terminating processes
- ✧ Prevention: Avoiding deadlocks by breaking one of the necessary conditions
- ✧ Total Ordering: Allocating resources in a fixed order to prevent circular wait

Output/Result:


CPU INSTRUCTIONS IN MEMORY (RAM)
SPECIAL CPU REGISTERS
GENERAL PURPOSE CPU REGISTERS
PROGRAM LIST
PROGRAM STACK (RAM)


RUNNING PROCESSES
SCHEDULER
READY PROCESSES (Ready
WAITING PROCESSES (Waiting Queue)
DATA MEMORY



Conclusion: The experiment demonstrated the concepts of memory management using virtual memory and paging, showing efficient memory utilization. It also illustrated the occurrence of deadlocks and the use of detection, prevention, and recovery techniques. Overall, the experiment helped in understanding how operating systems manage memory and handle deadlocks effectively.